

Sleep, Stress and Occupation:

An Unsupervised Analysis of the Sleep Health
and Lifestyle Dataset

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Abstract

We analyse the *Sleep Health and Lifestyle* dataset (374 synthetic records, 13 attributes) without ever showing the diagnostic label (Sleep Disorder) to any algorithm, asking whether the data organises itself into groups on its own and whether those groups have anything to do with a diagnosis they never saw. Three clustering techniques, agglomerative hierarchical clustering (Complete and Ward linkage), k -means (random and k -means++ initialisation) and DBSCAN, are each run on the full 21-dimensional feature matrix and on an 8-component PCA space, for ten configurations in total, compared through internal criteria (silhouette, WSS/BSS) and post-hoc external criteria (ARI, V-measure, purity). A Bayesian Network is then learned over the same variables plus the Sleep Disorder variable used as a node to model conditional dependencies. Even though no algorithm ever saw the Sleep Disorder label during fitting, agreement with it is well above chance across every configuration (ARI up to 0.224, purity up to 0.93), though never strong enough to reproduce the diagnosis outright. The Bayesian Network recovers a plausible physiological chain (BMI $\rightarrow$ sleep quality $\rightarrow$ sleep duration $\rightarrow$ stress $\rightarrow$ heart rate) and shows that stress and sleep quality inform the diagnosis only *indirectly*, through systolic pressure. A constraint-based structure (PC) agrees with the score-based one on very little ($F_1 = 0.122$), so the specific edges around Sleep Disorder are treated as hypotheses.

1 Introduction

Sleep is discussed constantly in the context of health, yet it remains one of the aspects most poorly captured by the kind of structured, tabular data typically used in machine learning. Most existing work is either purely clinical or purely supervised: a model is trained to predict a diagnosis from a fixed set of features. This project takes a different route. Rather than treating the sleep-disorder labels already present in the data as something to predict, we set them aside until the very end and ask a more open question: *does the data organise itself into groups on its own, and if it does, do those groups have anything to do with a diagnosis the algorithm never saw?*

The dataset is the *Sleep Health and Lifestyle Dataset* [1], a synthetic collection of 374 records and 13 attributes available on Kaggle. It combines demo-

graphics (age, gender, occupation), behavioural indicators (physical activity, stress level, daily steps), physiological measurements (BMI, blood pressure, heart rate) and sleep-related variables (duration, perceived quality, and a diagnosed disorder among *None*, *Insomnia* and *Sleep Apnea*). Because the data is synthetic, none of the conclusions below should be read as validated clinical findings.

Section 2 covers preprocessing; Section 3 the methodology and the model-selection criteria; Section 4 the clustering results and the selection of the final partition; Section 5 the Bayesian Network; Section 6 the joint reading of the two and the limitations.

2 Data and Preprocessing

Variable types. Person ID, Gender, Occupation and Sleep Disorder are nominal. Quality of Sleep, Stress Level and BMI Category are ordinal. Age, Sleep Duration, Physical Activity Level, Heart Rate and Daily Steps are ratio variables. Blood Pressure was the exception: stored as a single string (e.g. "126/83"), it packs two physiologically distinct quantities into one non-numeric column, and was split into Systolic Pressure and Diastolic Pressure.

Missing values. Only Sleep Disorder contains NaNs, and they are not missing in the usual sense: they correspond to subjects with no diagnosed condition. This is a Missing-Not-At-Random situation in which the absence is itself informative, so the NaNs were replaced with the explicit category *None*.

Encoding. Gender (binary) was mapped to a single 0/1 column. Occupation (11 unordered levels) was one-hot encoded, since arbitrary integers would imply a false notion of distance between professions. BMI Category has a real medical ordering and was ordinally encoded, after merging the duplicated labels *Normal* and *Normal Weight*. Quality of Sleep (4–9) and Stress Level (3–8) are six-level integer scales with roughly even spacing and were treated as approximately continuous, a choice validated by comparing Pearson and Spearman correlations against the other features: the largest gap found was ≈ 0.15 on Stress Level. Given how coarse a six-point scale already is, this seemed acceptable

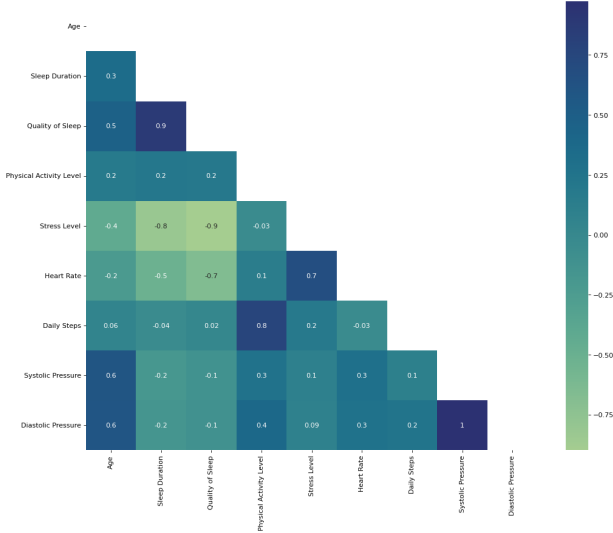


Figure 1: Correlation matrix of the numeric features. The stress/sleep/heart-rate block and the perfect systolic–diastolic correlation are the two structures that drive the feature-selection decisions.

Correlation and redundancy. Systolic and Diastolic Pressure turned out to be perfectly correlated ($\rho = 1.0$) in this dataset, so Diastolic Pressure was dropped as fully redundant (Fig. 1). Quality of Sleep, Sleep Duration, Stress Level and Heart Rate move together ($|\rho| \approx 0.7\text{--}0.9$), suggesting a single underlying physiological dynamic; Physical Activity Level and Daily Steps likewise ($\rho \approx 0.8$). Age stands apart, correlating moderately with blood pressure (≈ 0.6) and only weakly with sleep, stress or activity. The final feature matrix has 21 columns.

Outlier detection. Two methods resting on different assumptions were used. *Grubbs’ test* is univariate and iterative: for each feature it compares

$$G = \frac{\max_i |x_i - \bar{x}|}{s} \quad (1)$$

against a critical value derived from the t distribution, removes the flagged point and repeats until nothing crosses the threshold.

Local Outlier Factor instead compares the local density of a point with that of its k nearest neighbours $N_k(p)$. For each neighbour o , the *reachability distance* of p from o is

$$\text{reach-dist}_k(p, o) = \max\{d_k(o), d(p, o)\}, \quad (2)$$

where $d_k(o)$ is the distance from o to its own k -th nearest neighbour. The *local reachability density* of p is the inverse of the average reachability distance over its neighbourhood, and the LOF score is the ratio between the average density of the neighbours and the density of p itself,

$$\text{LOF}_k(p) = \frac{\frac{1}{|N_k(p)|} \sum_{o \in N_k(p)} \text{lr}_d(o)}{\text{lr}_d(p)}, \quad (3)$$

so a score well above 1 marks a point noticeably sparser than its surroundings, catching multivariate anomalies invisible to a marginal test. At $\alpha = 0.05$ Grubbs flagged 2 points (both on **Heart Rate**); LOF flagged 19 (5.1%, which is essentially the **contamination** parameter by construction). The two agreed only on the 2 points Grubbs found (Fig. 2). Since **contamination** forces LOF to flag roughly that fraction whether or not a real density gap exists, only the 2 points confirmed by both were removed, leaving **372 of 374 observations**.

Scaling. Every feature was scaled with **MinMaxScaler**,

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (4)$$

rather than standardised: several columns are already binary 0/1 one-hot indicators, and centring them around a mean and standard deviation carries no meaningful interpretation. Min–max scaling brings every column into $[0, 1]$ while leaving the bounded binary columns essentially untouched, which is what keeps each feature contributing comparably to the Euclidean distances used downstream.

3 Methodology

Three clustering techniques were applied, each on the full 21-dimensional feature space and on a PCA-reduced version, for ten configurations overall. The three were picked to cover distinct paradigms: hierarchical clustering builds a full nested structure without fixing k upfront, k -means partitions the space around centroids under a fixed k , and DBSCAN defines clusters by density and can leave points unassigned. Running all three made it possible to check whether any structure found held up across different notions of what a cluster is, or was simply an artefact of one algorithm’s assumptions.

3.1 Principal Component Analysis

PCA finds an orthogonal basis ordered by captured variance: the first component solves $w_1 = \arg \max_{\|w\|=1} \text{Var}(Xw)$ and each subsequent one maximises the remaining variance subject to orthogonality, which amounts to the eigendecomposition of the covariance matrix, or equivalently the SVD $X = U\Sigma V^T$. It is used here to remove redundant dimensions, to mitigate the curse of dimensionality affecting distance-based clustering, and to make visualisation possible (t-SNE was also used for this purpose, especially when plotting clustering labels). The number of components was chosen as the smallest covering at least 90% of total variance.

3.2 Hierarchical clustering

Agglomerative clustering starts with every point in its own cluster and merges the two closest at each step,

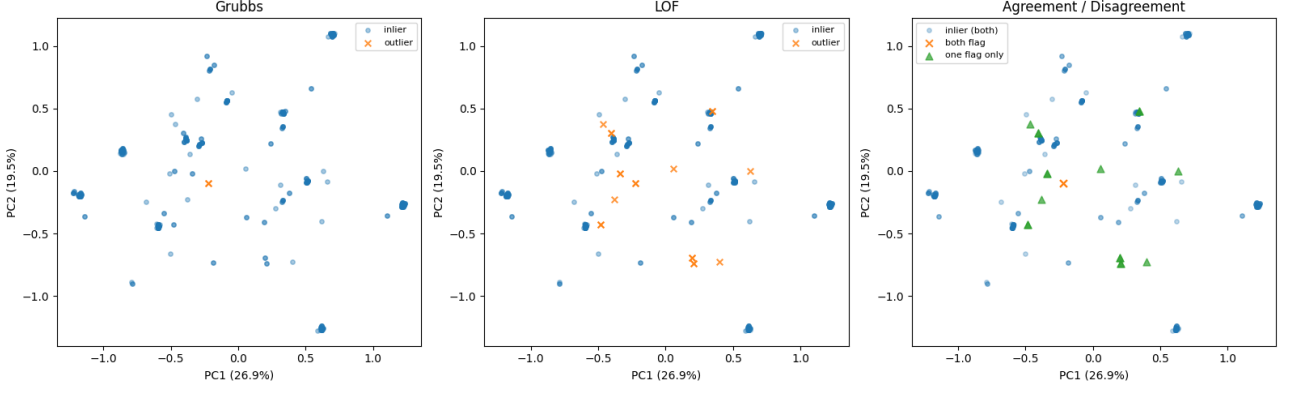


Figure 2: Grubbs vs. LOF outliers projected on the first two principal components, and their agreement. Only the two points flagged by both methods were removed.

recording the sequence in a linkage matrix drawable as a dendrogram. Two linkage criteria were compared. Unlike k -means, the merge sequence here is entirely deterministic: at every step the algorithm picks the single closest pair of clusters under the chosen linkage, so there is no local-minimum risk and nothing to stabilise against random initialisation. *Complete linkage* takes the largest pairwise distance between members,

$$d(A, B) = \max\{d(a, b) : a \in A, b \in B\}, \quad (5)$$

a conservative rule that yields compact, similarly sized clusters but is sensitive to outliers, since one distant point inflates the inter-cluster distance. *Ward linkage* merges the pair causing the smallest increase in total within-cluster sum of squares,

$$\Delta\text{WSS}(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2, \quad (6)$$

producing balanced, compact clusters, at the cost of assuming convex, roughly isotropic shapes.

Because the merge criterion is built on squared distances to a cluster mean, the same way k -means is, Ward inherits some of that sensitivity to outliers: a single far point pulls the relevant centroid and inflates the variance the criterion is trying to minimise. The effect is usually milder than Complete linkage’s, since it is diluted across the whole cluster.

3.3 K-means

K-means alternates assignment to the nearest centroid and recomputation of centroids as cluster means, minimising

$$\text{WSS} = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2. \quad (7)$$

WSS decreases monotonically, so convergence is guaranteed, but only to a local minimum depending on initialisation. It was run with 50 restarts, keeping the lowest-WSS solution. Two initialisations were compared: random placement, and k -means++, which seeds each new centroid with probability proportional to its squared distance from the nearest chosen one

and carries an $O(\log k)$ approximation guarantee on the final WSS.

Because centroids are plain means, k -means is sensitive to outliers for the same reason Ward is: one extreme point shifts its centroid and distorts every distance computed from it. This is part of why outlier removal was handled as a preprocessing step.

3.4 DBSCAN

DBSCAN [2] defines clusters as dense regions separated by sparser ones. With $N_\varepsilon(p) = \{q : d(p, q) \leq \varepsilon\}$, a point is *core* if $|N_\varepsilon(p)| \geq \text{MinPts}$, *border* if it lies in the neighbourhood of a core point without being core, and *noise* otherwise. This lets DBSCAN recover arbitrarily shaped clusters and explicitly flag outliers, something neither k -means nor hierarchical clustering can do, at the price of high sensitivity to ε and MinPts. ε was first estimated geometrically with the k -distance knee method, then refined by a grid search in a narrow window around that estimate. Because silhouette computed on non-noise points rewards configurations that simply discard more points, among configurations within a small tolerance of the best score the one with the lowest noise fraction was preferred over the raw argmax. Once ε and MinPts are fixed, the assignment step itself is deterministic: there is no random initialisation and no local-minimum risk to control for. Whatever variability shows up across runs traces back to how those two hyperparameters were chosen, not to the algorithm’s convergence behaviour.

3.5 Model-selection criteria

Ten configurations mean weighting criteria that measure different, partly conflicting things, and no single number is treated as sufficient. The **internal** criterion is the silhouette,

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}, \quad \bar{s} = \frac{1}{n} \sum_{i=1}^n s(i), \quad (8)$$

with $a(i)$ the mean distance to the other points of its own cluster and $b(i)$ the mean distance to those of the

nearest other cluster, tracked alongside the variance decomposition $BSS = TSS - WSS$. The **external** criteria are the Adjusted Rand Index, defined as the pairwise Rand Index corrected for chance agreement, so that a random partition scores ≈ 0 and the V-measure, the harmonic mean of homogeneity h and completeness c ,

$$h = 1 - \frac{H(C | K)}{H(C)}, \quad c = 1 - \frac{H(K | C)}{H(K)}, \quad V = \frac{2hc}{h+c}, \quad (9)$$

computed against the **Sleep Disorder** label that no algorithm saw during fitting; purity, Jaccard and Fowlkes–Mallows were tracked alongside them.

The internal and external rankings are read side by side, under two structural constraints. First, **coverage**: DBSCAN’s silhouette is computed only on the points it assigns, so it is not comparable with that of a method classifying every point, and a partition leaving a fifth of the subjects unlabelled is of limited use for downstream interpretation. Second, **cluster-size balance**: a silhouette propped up by near-singleton clusters reflects easy points being carved off, not a better partition. For this reason k was chosen for every method by combining the silhouette curve and the WSS/BSS elbow with an explicit inspection of cluster sizes across a range of k , taking the largest k before clusters of a handful of points start appearing.

4 Clustering Results

4.1 Full feature space

Projecting the scaled data on its first two principal components (Fig. 3) already shows that the first two components capture only 46.8% of total variance (27.1% and 19.7%) and that the three diagnostic classes overlap heavily. This is expected going into an unsupervised analysis: were the classes cleanly separated, clustering would merely reproduce the labels.

Hierarchical clustering. For Complete linkage, cluster sizes stay reasonable up to $k = 11$, but from $k = 12$ near-singleton clusters of 2 points appear which is exactly what drives the rise in silhouette at higher k . We therefore take $k = 10$, the last configuration before fragmentation. Ward keeps sizes reasonable from $k = 6$ to $k = 11$, with the balance breaking sharply at $k = 12$, so $k = 11$ is the natural cutoff (Fig. 4, 5). Ward reaches a silhouette of 0.736 against Complete’s 0.593, with a much lower WSS (89.71 vs. 157.50) and higher BSS (583.98 vs. 516.19), consistent with Ward minimising variance directly while Complete only bounds the maximum pairwise distance. Externally the two sit close: Ward wins on purity, homogeneity, completeness and V-measure, while Complete is marginally ahead on ARI and Fowlkes–Mallows. This happens because more, smaller clusters leave fewer opportunities for points of different diagnoses to co-occur, which inflates pair-

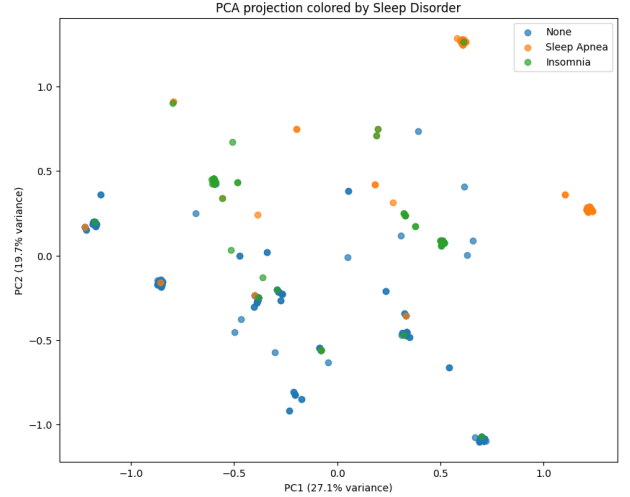


Figure 3: PCA projection coloured by **Sleep Disorder**. The three classes overlap in the plane of maximum linear variance, which is precisely what makes the unsupervised question non-trivial.

based scores without necessarily meaning the labels are better captured (Table 1).

K-means. Random initialisation gives an irregular size pattern as k grows (balanced up to $k = 9$, a sharp drop at $k = 10$, partial recovery at $k = 11$, collapse at $k = 12$), so $k = 11$ was taken as a compromise; k -means++ behaves more predictably and stays balanced up to $k = 10$, which is its natural cutoff. Internally the two land at 0.717 and 0.708, both below Ward, despite the extra restarts, and their sorted similarity matrices are visibly more uneven than Ward’s (Fig. 6).

DBSCAN. ε was first estimated geometrically with the k -distance knee method (Fig. 7), giving $\varepsilon \approx 0.073$ at MinPts = 15; a grid search in a narrow window around that estimate then settled on the conservative pick $\varepsilon = 0.351$, MinPts = 16. The resulting partition has 11 clusters with 79 points (21.2%) labelled noise. The resulting silhouette, 0.962, is computed only on the 293 assigned points and is therefore *not* comparable with the values above. On external metrics, however, DBSCAN clearly leads the full-feature comparison (ARI 0.183, purity 0.928, homogeneity 0.718, V-measure 0.415; Table 1). The t-SNE projection in fig. 8 shows the noise points scattered along the boundaries between clusters, consistent with points sitting in low-density transition zones.

4.2 PCA-reduced space

The cumulative explained-variance curve flattens gradually with no sharp elbow: 8 components are needed for 90% of the variance (10 for 95%), as shown in fig. 9, so the dimensionality was cut from 21 to 8. The loadings, in fig. 10, show the structure of the PCA space. PC1 is dominated by **Gender** and, less strongly, **Age** and **Stress Level**; PC2 mixes sleep

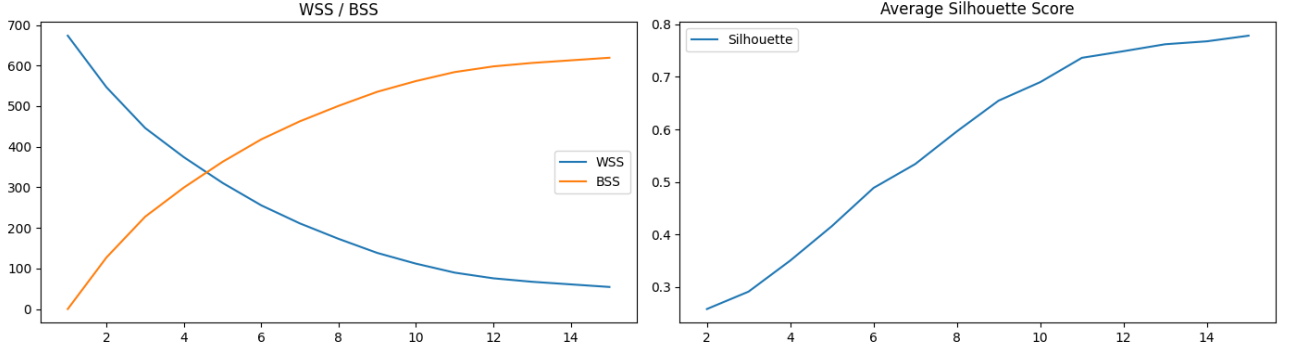


Figure 4: WSS/BSS and average silhouette across $k = 1$ –15 for Ward linkage on the full feature space. WSS decreases and BSS increases smoothly with k ; silhouette rises gradually to $k = 11$ before climbing further at higher k , a range excluded here because cluster sizes fragment past that point.

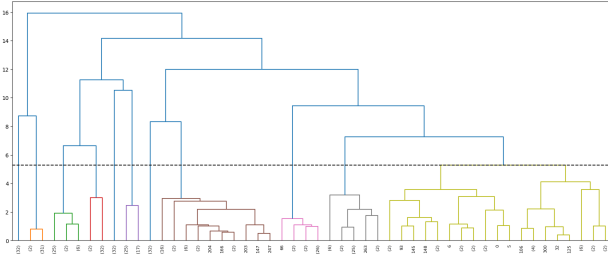


Figure 5: Ward dendrogram on the full feature space, truncated to the last 50 merges and cut at $k = 11$ (height ≈ 5.3). Several branches stay distinct well past the cut, matching the balanced cluster sizes.

and lifestyle variables, contrasting high-stress/poor-sleep profiles against low-stress ones; PC3 is essentially an activity axis. From PC4 onward the components become occupation-specific, each loading mainly on one or two one-hot occupation columns. The one-hot encoding introduces a set of weakly correlated directions that PCA can only compress so far. Complete linkage is the method that benefits most from the reduction: its k revises down to 8 (a 2-point cluster appears from $k = 9$ onward) and *every* external metric improves: ARI from 0.152 to 0.224, V-measure from 0.283 to 0.352, purity from 0.788 to 0.863. This matches what the loadings suggest: since the components beyond the third mostly encode individual occupations, compressing them removes noise from the distance computations, and Complete linkage was the method most penalised by that noise. Ward, conversely, barely changes externally (ARI 0.140, V-measure 0.313) and its silhouette does rise to 0.770, the best of any configuration classifying every point. Both k -means variants settle on $k = 10$ with clean size profiles and improved silhouettes (0.757 and 0.760) but essentially unchanged external metrics.

DBSCAN reverses completely. On the reduced space the knee-based grid search settles on a much coarser radius, only 3 clusters and almost no noise (2 points), but one cluster absorbs 306 of the 372 points, with the other two at 32 each. This is one dominant group plus two small pockets and every external metric collapses

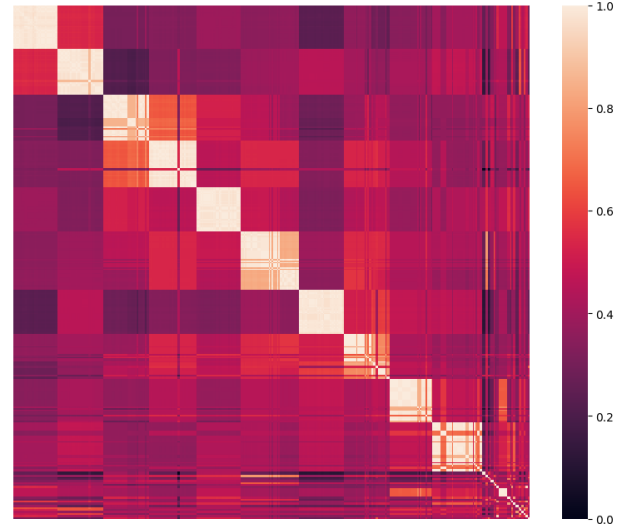


Figure 6: Sorted similarity matrix for Ward ($k = 11$). Dense blocks along the diagonal indicate that points inside a cluster are mutually similar; the residual bleeding between adjacent blocks marks the clusters that sit in overlapping regions.

accordingly (ARI 0.066, homogeneity 0.150, purity 0.665; Table 1).

4.3 Model selection

Internally, and restricting attention to the eight configurations that classify every point (Section 3.5; Table 1), Ward (PCA) leads with a silhouette of 0.770 and the best variance decomposition of the group, followed by k -means++ (PCA) at 0.760 and Ward on the full space at 0.736; Complete linkage sits last on both spaces (0.59). DBSCAN’s 0.962 on the full space is excluded from this comparison because it evaluates only the easiest 79% of the data.

Externally, Complete linkage (PCA) leads (ARI 0.224, V-measure 0.352), the largest improvement of any method between the two feature spaces, with full-space DBSCAN second on ARI (0.183) and first of all on V-measure (0.415). The remaining seven configurations cluster tightly, ARI between 0.133 and 0.152,

Table 1: Internal and external metrics for all ten configurations: full feature space (372 points) and PCA-reduced space (8 components).

Feature space	Method	k	Noise	Sil.	ARI	Pur.	V
Full (372 pts)	Hier. Complete	10	0	0.593	0.152	0.788	0.283
	Hier. Ward	11	0	0.736	0.140	0.855	0.332
	K-Means (random)	11	0	0.717	0.144	0.863	0.324
	K-Means++	10	0	0.708	0.133	0.841	0.297
	DBSCAN	11	79	0.962 [†]	0.183	0.928	0.415
PCA (8 comp.)	Hier. Complete	8	0	0.595	0.224	0.863	0.352
	Hier. Ward	11	0	0.770	0.140	0.858	0.313
	K-Means (random)	10	0	0.757	0.133	0.841	0.298
	K-Means++	10	0	0.760	0.138	0.850	0.312
	DBSCAN	3	2	0.244	0.066	0.665	0.187

[†] computed on the 293 non-noise points only; not comparable with the other rows.

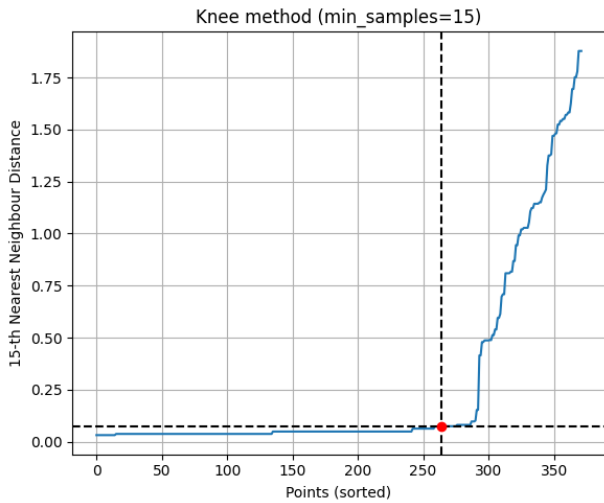


Figure 7: k -distance knee method for DBSCAN on the full feature space (MinPts = 15). The knee marks the geometric candidate $\varepsilon \approx 0.073$, later refined to $\varepsilon = 0.351$ by the narrow grid search.

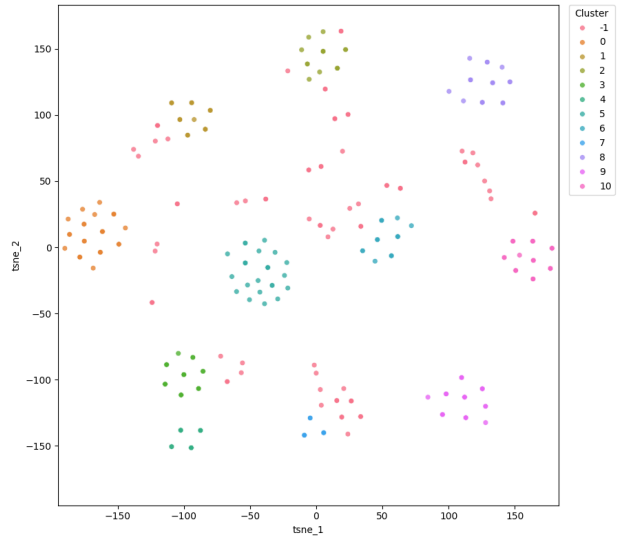


Figure 8: t-SNE projection of the DBSCAN partition on the full feature space. Noise points (label -1) lie at the boundaries between dense regions.

with no decisive advantage among them.

Weighing the two rankings under the coverage and size-balance constraints, we carry forward **Ward on the PCA-reduced space** ($k = 11$):

- it is the strongest configuration on internal grounds among those assigning all 372 subjects, and its cluster sizes are the cleanest of anything tested, with no tiny clusters (clusters with less than three members) appearing even at $k = 15$, so its silhouette is earned by the partition as a whole;
- its external metrics (ARI 0.140, V-measure 0.313) are modest, but they sit inside the band where seven of the ten configurations live;
- the two configurations that beat it externally each carry a structural cost that would propagate into every downstream interpretation: DBSCAN leaves 21% of subjects unassigned, so any profile built on it silently describes only part of the population, while Complete linkage has the worst internal geometry of the partition-based methods on either feature space.

5 Bayesian Network

The second half of the analysis changes formalism entirely: instead of grouping observations, it models the conditional dependencies *between variables* through a Directed Acyclic Graph in which each node is a variable and each edge a direct probabilistic influence. If the joint distribution is Markov with respect to the DAG it factorises as

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i)), \quad (10)$$

which is what makes the problem tractable at all: the full joint table over the 22 variables used here would be hopeless to estimate directly. Such a graph encodes *statistical dependence*; reading an edge causally would require assuming every parent is a direct cause of its children, an assumption we do not make and which a few of the learned edges would plainly violate.

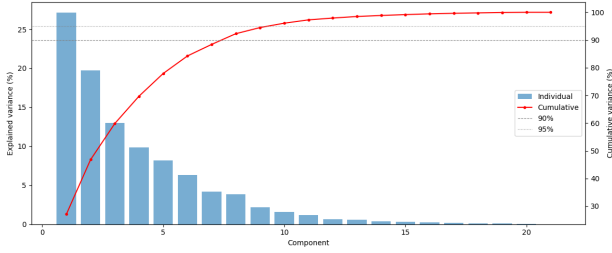


Figure 9: Individual and cumulative explained variance. Eight components are retained (90% of total variance).

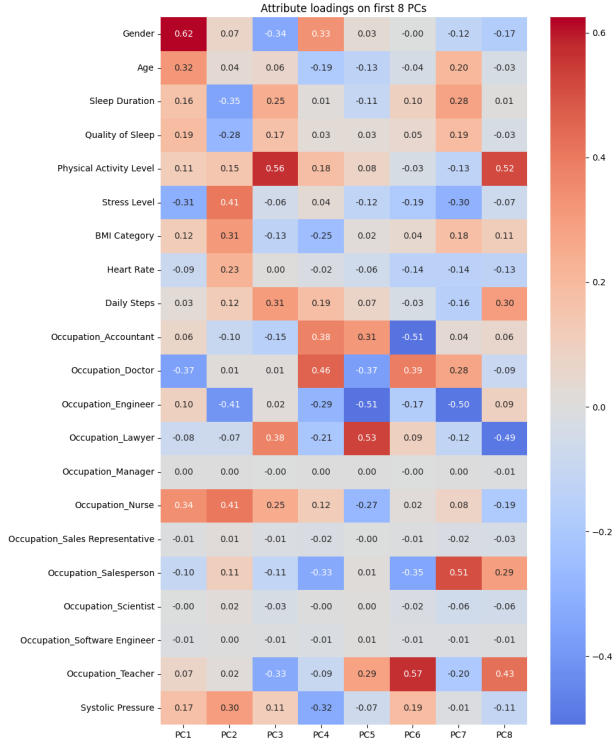


Figure 10: Attribute loadings on the first 8 principal components. From PC4 onward the components mostly isolate individual occupations.

5.1 Discretisation and structure learning

Both BIC scoring and `pgmpy`'s parameter estimation require discrete data, so the eight continuous variables were discretised into tertiles by quantile-based binning. Quantile binning was preferred over fixed-width binning because, for skewed variables, fixed-width bins can be left nearly empty, and poorly populated bins translate directly into unreliable conditional probability estimates. Binary and already-discrete columns were left untouched.

The structure was learned with Hill Climbing, a greedy score-based search starting from an empty graph, applying at each step whichever single edge addition, removal or reversal most improves the BIC score

$$\text{BIC}(G, D) = \log P(D | \hat{\theta}_G, G) - \frac{\log N}{2} |\theta_G|. \quad (11)$$

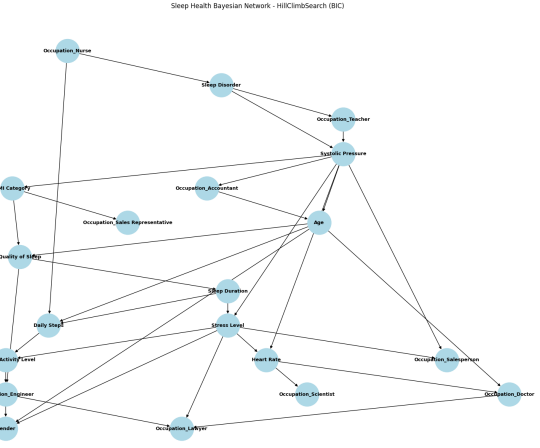


Figure 11: DAG learned by Hill Climbing with the BIC score (34 edges, 2 isolated nodes). **Systolic Pressure**, **Age** and **Stress Level** act as hubs; **Sleep Disorder** is peripheral.

The log-likelihood term always rewards adding parents, since a richer conditional table can only fit the data at least as well; the penalty grows quickly as a node accumulates parents, because table size is multiplicative in parent cardinalities. An edge therefore survives only if its explanatory gain justifies its complexity, which is what keeps the learned graph sparse. The resulting DAG has 34 edges out of the 462 directed edges possible over 22 nodes (Fig. 11). Two occupation columns, **Occupation_Software Engineer** and **Occupation_Manager**, end up with no edges at all, meaning the search found no dependency involving either category strong enough to justify the added model complexity; every other variable retains at least one connection. **Systolic Pressure**, **Age** and **Stress Level** emerge as hubs. The lower part of the graph reproduces a coherent physiological chain: **BMI Category** → **Quality of Sleep** → **Sleep Duration** → **Stress Level** → **Heart Rate**, matching the correlation analysis. **Sleep Disorder** itself is peripheral, with a single parent (**Occupation_Nurse**) and two children (**Occupation_Teacher**, **Systolic Pressure**). Not every edge is defensible on domain grounds: **Age** sending an edge to **Gender**, and receiving one from **Occupation_Accountant**, is statistically supported but implausible if read causally, an expected limitation of a search that optimises fit with no notion of temporal precedence, and one that Markov-equivalent DAGs cannot resolve from observational data alone.

5.2 Markov blanket and inference

The CPDs were fitted by Maximum Likelihood, and the model passes all internal consistency checks. The Markov blanket of **Sleep Disorder**,

$$\text{MB}(X) = \text{Pa}(X) \cup \text{Ch}(X) \cup \bigcup_{Y \in \text{Ch}(X)} \text{Pa}(Y) \setminus \{X\}, \quad (12)$$

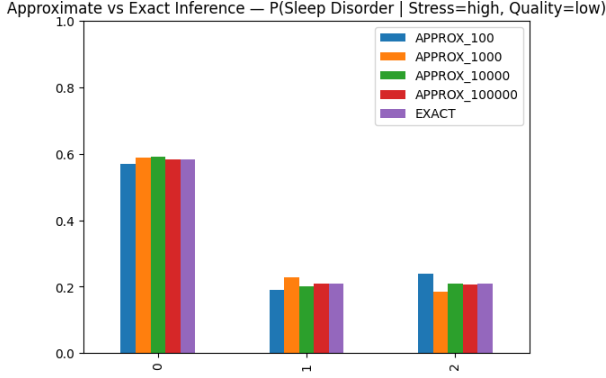


Figure 12: Approximate (forward sampling, 10^2 – 10^5 samples) vs. exact inference for $P(\text{Sleep Disorder} \mid \text{Stress} = \text{high}, \text{Quality} = \text{low})$.

turns out to contain only three variables: `Occupation_Nurse`, `Systolic Pressure` and `Occupation_Teacher`. Notably the latter appears in a double role, as a direct child of `Sleep Disorder` and as a co-parent of `Systolic Pressure`, so the blanket stays at three elements because the only one that exists is already in the set.

Neither `Stress Level` nor `Quality of Sleep` belongs to the blanket, which initially suggested that conditioning on them should leave $P(\text{Sleep Disorder})$ unchanged. Exact inference (`VariableElimination`) refutes this: $P(\cdot \mid \text{Stress} = \text{high}, \text{Quality} = \text{low})$ gives (58.3%, 20.9%, 20.8%) over (None, Apnea, Insomnia), while $P(\cdot \mid \text{Stress} = \text{low}, \text{Quality} = \text{high})$ gives (45.8%, 36.6%, 17.6%).

The resolution lies in the exact statement of the blanket property: it guarantees conditional independence *only when the entire blanket is observed*, and says nothing about marginal independence when it is not. Here `Systolic Pressure`, a child of `Sleep Disorder` and a member of the blanket, is *not* part of the evidence, so the paths running through it stay active and the information observed on the evidence variables propagates back to `Sleep Disorder`, exactly as observing an effect lets one update belief about its cause. The mechanism differs between the two evidence variables: one reaches the target along an unblocked chain, the other only *because* conditioning on `Stress Level`, a collider on that path, unblocks it. The practical lesson is that the relevance of a variable to a query cannot be read off the Markov blanket alone: d-separation must be checked against the actual evidence set, including whether any collider on the path happens to be part of it.

This result was cross-checked with forward sampling (`ApproxInference`) at increasing sample sizes, from 10^2 to 10^5 . The approximate estimates converge steadily to the exact values, landing within half a percentage point at 10^5 samples for both queries (Fig. 12).

5.3 Comparison with the PC algorithm

To gauge how much of the learned structure is an artefact of greedy search, a second structure was learned with the PC algorithm, a constraint-based method. Where Hill Climbing asks which graph scores best overall, PC repeatedly asks whether two variables are conditionally independent given some subset of the others: it prunes a fully connected skeleton with χ^2 tests at $\alpha = 0.05$, then orients edges from the pattern of colliders found. It is far more conservative by construction: an edge survives only if a statistical test actively rejects independence.

PC returns 15 edges against Hill Climbing’s 34, leaving six variables completely isolated

(`Occupation_Software Engineer`, `Occupation_Manager`, `Occupation_Scientist`, `Occupation_Nurse`, `Occupation_Doctor`, `Occupation_Sales Representative`) and eight more with only a single edge (Fig. 13). The two structures overlap on little: only three directed edges coincide between them (`Occupation_Engineer` → `Occupation_Lawyer`, `Sleep Duration` → `Stress Level`, `Daily Steps` → `Physical Activity Level`). The stress/sleep cluster surfaces in both graphs, though two of the shared relationships come out reversed between the two methods (`Quality of Sleep` vs. `Sleep Duration`, and `Physical Activity Level` vs. `Stress Level`); Hill Climbing’s hub, `Systolic Pressure`, barely registers in PC’s graph, and the demographically implausible directions do not appear. Most relevant here, `Sleep Disorder` again has a single edge, but from `Occupation_Teacher` rather than `Occupation_Nurse`, the two methods agree that *some* occupation variable drives the diagnosis in this discretised representation, and disagree on which.

Agreement measured as an F_1 score over directed edges,

$$F_1 = \frac{2 |E_{\text{HC}} \cap E_{\text{PC}}|}{|E_{\text{HC}}| + |E_{\text{PC}}|} = \frac{2 \times 3}{34 + 15}, \quad (13)$$

comes out at 0.122. This is not a failure of either method: PC’s stricter threshold can miss real but weak dependencies, especially with 374 observations spread across discretised bins, while BIC can retain edges a stricter test would reject. It does mean, however, that a non-negligible share of Hill Climbing’s edges, including the `Occupation_Nurse` → `Sleep Disorder` edge underpinning the inference discussion above, should be treated as a hypothesis suggested by one search procedure.

6 Discussion and Conclusion

The two halves of the analysis tell a consistent, if partial, story. Even without the algorithm ever seeing a label, agreement with the `Sleep Disorder` label is well above chance across every configuration (ARI up to 0.224, purity up to 0.93), though never strong enough

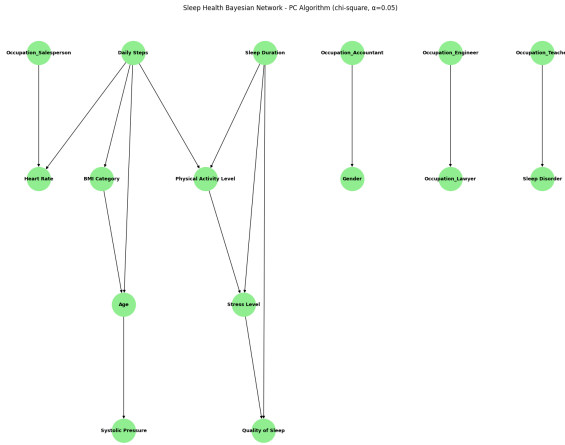


Figure 13: DAG learned by the PC algorithm (15 edges, $\alpha = 0.05$). Considerably sparser than the Hill Climbing structure, with six isolated nodes.

to reproduce the diagnosis outright. The Bayesian Network adds a variable-level view: it recovers a plausible physiological chain from BMI through sleep quality, sleep duration, stress and heart rate, and shows that stress and sleep quality do carry information about the diagnosis, but *indirectly*, through systolic pressure. Both techniques converge on the same reading from different angles: occupation-linked lifestyle and physiological patterns are more informative about sleep health in this dataset than any single variable in isolation, and the effect of stress and sleep quality on disorder risk is mediated.

Several limitations should be taken into account. The dataset is **synthetic**, so no figure reported here should be generalised to a real clinical population without validation. No clustering method fully recovered the three diagnostic categories: **completeness stayed the weakest external metric across every method and every feature space**, suggesting that healthy and pathological subjects genuinely overlap in this feature space. The selected model was chosen by balancing internal geometry, coverage and post-hoc label agreement.

On the network side, the low agreement between Hill Climbing and PC ($F_1 = 0.122$) means the specific edges around **Sleep Disorder** must be read as hypotheses.

Regarding **scalability**, the three techniques behave very differently at scale. Agglomerative clustering requires the full $n \times n$ dissimilarity matrix, hence $O(n^2)$ memory and $O(n^2 \log n)$ time, unusable beyond a few tens of thousands of points, and the clear loser for a real-time setting despite being the method selected here on a 372-point dataset. k -means runs in $O(nkdi)$ per restart and admits mini-batch and streaming variants, making it the most viable of the three for millions of points. DBSCAN sits in between: $O(n \log n)$ with a spatial index in low dimensions, but degrading toward $O(n^2)$ as dimensionality grows and the index loses effectiveness.

Future work could validate these findings on a real,

non-synthetic sleep dataset; inject domain knowledge into structure learning (for instance forcing **Age** and **Gender** to be root nodes, which would eliminate the implausible edges outright); or combine the two techniques explicitly, using the cluster assignment itself as an additional node in the network to see whether it mediates some of the relationships found here.

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